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# BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : TerraSynapse

Team Leader Name : Karthikeyan R

Problem Statement : Cross-Modal Satellite Image Retrieval Using Multi-Sensor Remote Sensing Data

## Team Members

Team Leader:

Name: Karthikeyan R

College: Vellore Institute Of Technology

Team Member-1:

Name: Rebecca J

College: Vellore Institute Of Technology

Team Member-2:

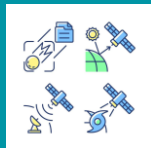
Name: Bharath I

College: Vellore Institute Of Technology

Team Member-3:

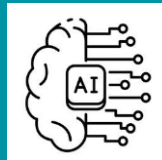
Name: Musharraf Hamdan K

College: Vellore Institute Of Technology



### How is it different?

- Integrates Optical, SAR, and Multispectral imagery in one framework.
- Uses dedicated modality-specific encoders.
- Learns a sensor-independent shared embedding space.
- Supports same-modal and cross-modal retrieval.



### How will it solve the problem?

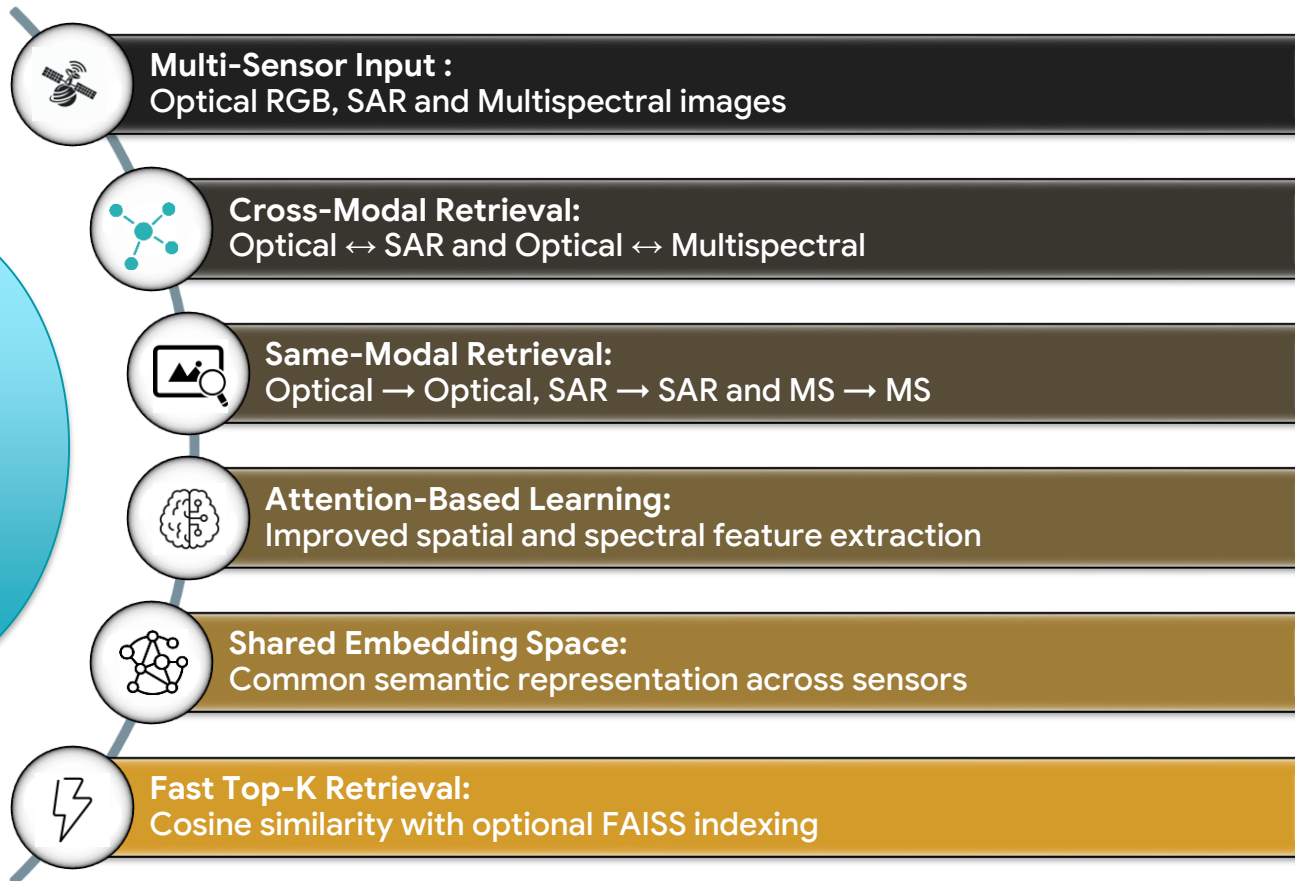
- Converts paired satellite images into aligned feature embeddings.
- Uses attention mechanisms to capture spatial and spectral information.
- Applies multi-task losses for better semantic alignment.
- Retrieves top-k similar images using cosine similarity or FAISS.



### Unique Selling Proposition

- One scalable framework for multiple satellite sensors.
- Reduces sensor-domain differences through shared semantic learning.
- Expandable to large satellite-image archives.

## List Of Features offered



## Data Acquisition

Paired Multi-Sensor Datasets

- SEN12MS-CR
- BIGEarthNet v2.0
- EuroSAT-SAR



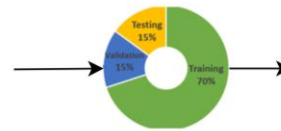
## Data Preprocessing

Example  
Class: AnnualCrop

Optical

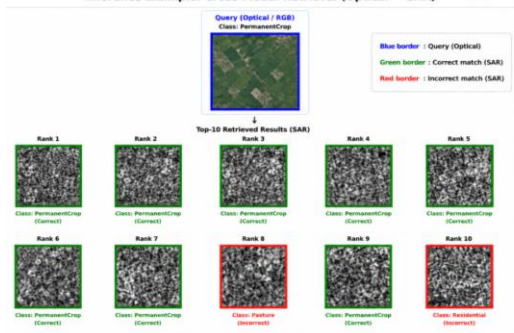
SAR

MultiSpectral

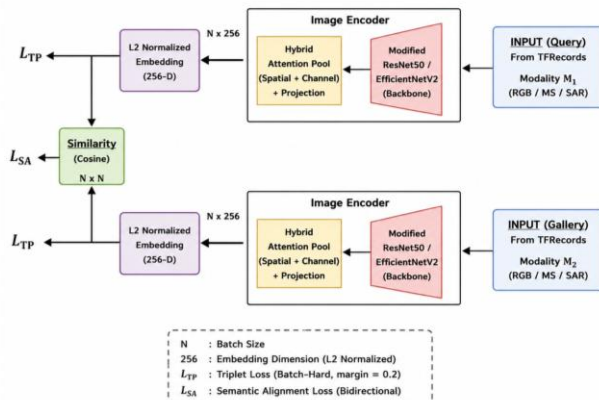


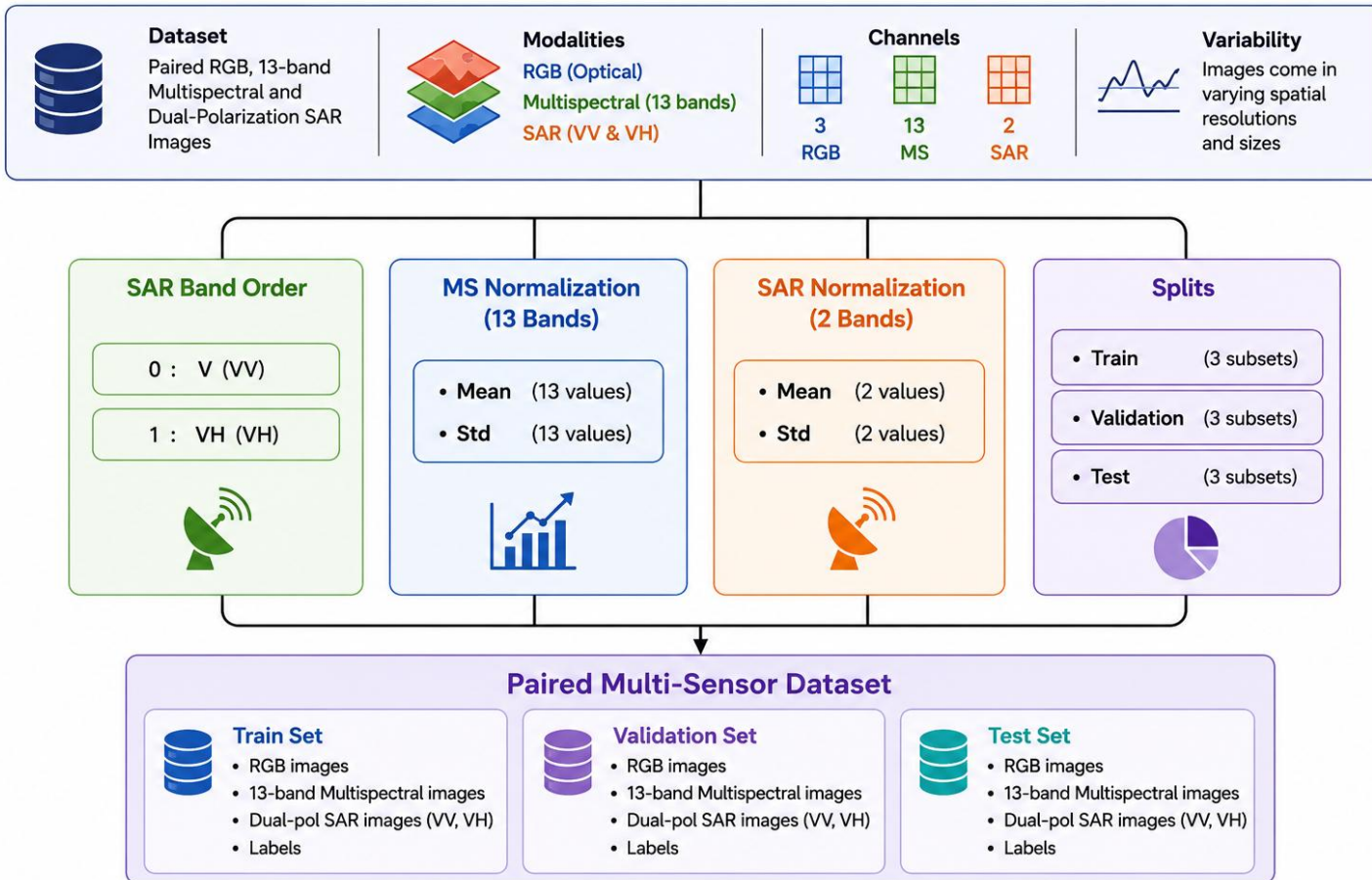
## Inference Examples

Inference Example: Cross-Modal Retrieval (Optical → SAR)

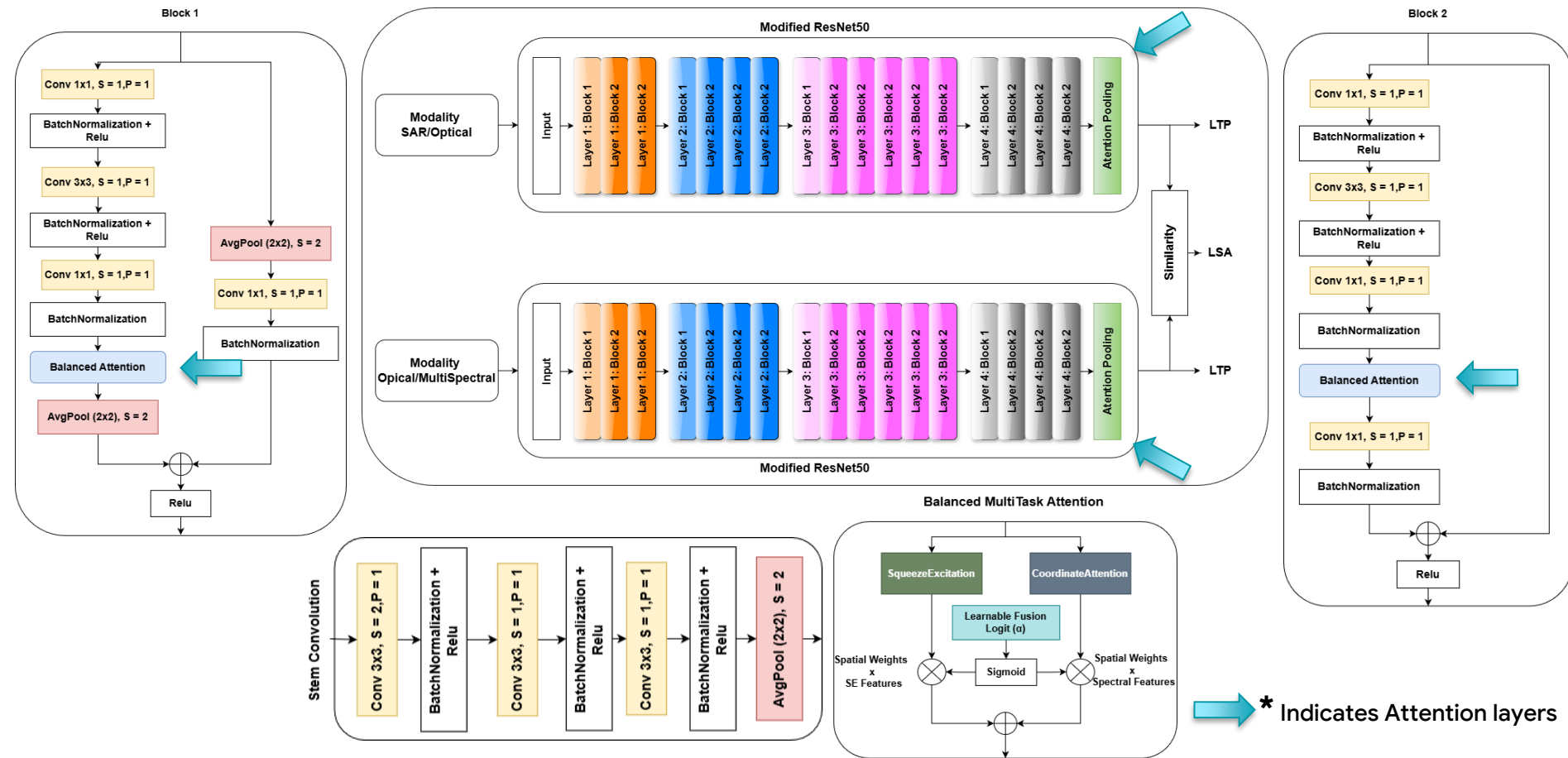


## Model Training

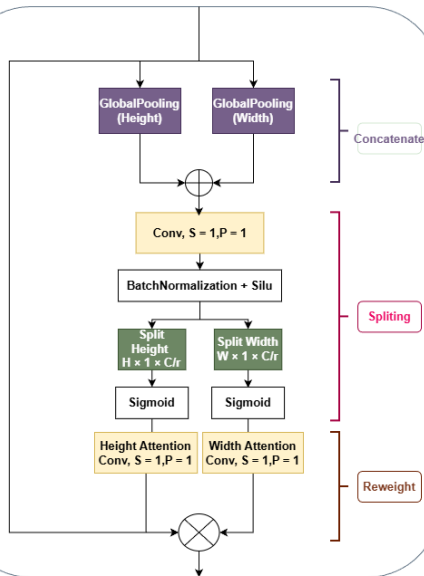




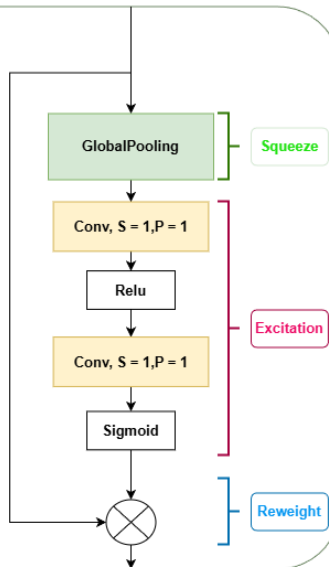
# Proposed ResNet-50 Architecture



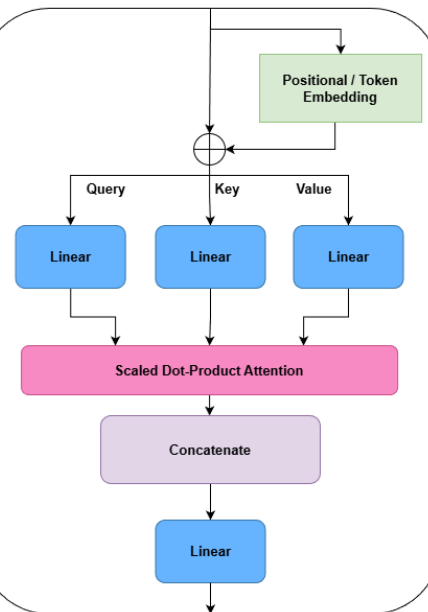
Coordinate Attention



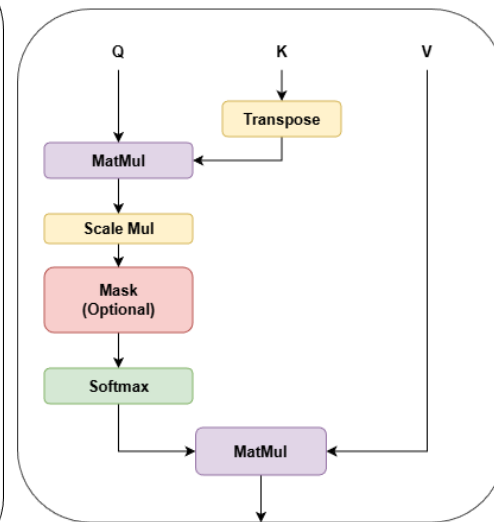
Squeeze Excitation



Attention Pooling



Scaled Dot-Product Attention





# Technology To Be Used



## Programming Language

Python 3.x for development and implementation



TensorFlow

## Deep Learning Framework

TensorFlow and Keras for model building and training



Keras

## High-level API

Keras for efficient model design and experimentation



ResNet50



EfficientNetV2

## Backbone Architectures

Modified ResNet50 and EfficientNetV2 as encoders



## Attention Mechanisms

Balanced Attention, Coordinate Attention, SE, Attention Pooling



TFRecord

## Data Pipeline

TFRecords and TensorFlow tf.data for fast and scalable data loading



NumPy

## Numerical Computing

NumPy for numerical operations and array manipulation



pandas

## Data Handling

Pandas for data preprocessing, cleaning and analysis



## Machine Learning Tools

Scikit-learn for evaluation metrics and utilities



## Image Processing

OpenCV for image preprocessing and augmentations



rasterio

## Geospatial Processing

Rasterio for reading and handling SAR and multispectral raster data

FAISS

## Similarity Search

FAISS for efficient similarity search and scalable top-k retrieval



matplotlib

## Data Visualization

Matplotlib / Seaborn for visualization of results and analysis



## Development Platform

Google Colab for development, training and experimentation

kaggle

## Computing Platform

Kaggle Notebook with GPU support for large-scale model training



## Dataset

EuroSAT (RGB), EuroSAT-SAR, EuroSAT-MS (Sentinel-2)



## Hardware

NVIDIA GPU (Kaggle/Colab) for accelerated training



## Storage

Local / Cloud storage for dataset and model artifacts



## Version Control

Git & GitHub for code versioning and collaboration



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HACKATHON

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THANK YOU